Practical 8: Simulation of AR(1) and MA(1) Processes

Question: Simulate an AR(1) model defined as $X_t = 0.6X_{t-1} + \epsilon_t$ and an MA(1) model. Plot both simulated series and explain the difference in behavior between AR and MA processes based on the graphs.

8.1 Objective

- To simulate an AutoRegressive (AR) process and a Moving Average (MA) process using Python.
- To visually compare the behavior and "memory" of these two fundamental time series models.

8.2 Theory

- **AR(1) Process ($X_t = \phi X_{t-1} + \epsilon_t$):** The current value depends on the *previous value* plus a shock. This creates a "long-term memory" effect, where shocks die out slowly over time.
- **MA(1) Process ($X_t = \epsilon_t + \theta\epsilon_{t-1}$):** The current value depends on the *current shock* and the *previous shock*. This creates a "short-term memory" effect, where a shock affects the series for only one or two periods before disappearing.

8.3 Code Implementation

```
# 1. Configuration np.random.seed(42) n = 200 phi = 0.6
# AR parameter theta = 0.6 # MA parameter epsilon = np.random.normal(0, 1, n)
# White noise # 2. Simulate AR(1) Process
# Formula: X_t = phi * X_{t-1} + epsilon_t ar1 = np.zeros(n) ar1[0] = epsilon[0] for t in
range(1, n): ar1[t] = phi * ar1[t-1] + epsilon[t]
# 3. Simulate MA(1) Process
# Formula: X_t = epsilon_t + theta * epsilon_{t-1} ma1 = np.zeros(n) ma1[0] = epsilon[0] for t
in range(1, n): ma1[t] = epsilon[t] + theta * epsilon[t-1]
```

8.4 Interpretation

- **AR(1) Behavior:** The graph is **smoother and wavier**. Because X_t depends on X_{t-1}, a high value tends to be followed by another high value. The effect of a shock "echoes" into the future, decaying slowly.
- **MA(1) Behavior:** The graph is **choppier and spikier**. Because X_t depends only on the current and previous error, the system "forgets" shocks very quickly. If a spike occurs, the series returns to the mean (zero) almost immediately.